

Category Discovery on Instamart: A Gap and an Opportunity

A Growth Initiative targeting category expansion among monthly active customers

CORE ISSUE:

Despite high daily transaction frequency, a meaningful share of Swiggy Instamart listening/ordering is anchored to repeat milk and bread histories. Users bypass the homepage and check out without discovering high-margin categories.

CURRENT GAPS:

- Pre-Checkout Fee Tax: Surcharges on small orders.
- Freshness Distrust: Hard produce return policies.
- Visual Noise: Flashing ads cause navigation anxiety.
- Specification Gaps: Lack of reviews/charts.

Platform	Discovery Feature	2026 AI Move
Instamart	Post-Checkout Add-on	Fee-free sandbox
Zepto	Pass discount widgets	Zepto Pass bundle
Blinkit	Gourmet Specialties	Regional local hubs
BigBasket	BB Daily Subscription	Consolidated deliveries

One Prioritized Problem: The Checkout Boundary Block

THE PROBLEM: Bypassing Homepage Discovery for Habitual Reorders

Our primary target segment (Convenience-First professionals, ages 22-35) and busy parents checkout items in under 15 seconds. They bypass homepage grids entirely, relying on 'Buy it Again' reorder lists for milk and bread. This anchors them to a small subset of the catalog and blocks high-margin category exploration.

Why they abandon auxiliary categories (The Core Friction):

- **The Delivery Fee Penalty:** If a user forgets a low-consideration item (e.g. eye drops, gourmet cheese) and remembers after checkout, ordering it separately adds ■35-50 in delivery and handling surcharges, which instantly kills the purchase.
- **Homepage Avoidance:** Users avoid browsing the homepage due to cognitive load from flashing promotional banner ads, missing the new catalog expansions.

Target Segment & Validated Pain Points through Research

Question Tested	Hypothesis	Val %	Verdict
Why struggle to discover?	Recommendations anchored to past histories.	81%	Strongly Validated
What are banner frustrations?	Promo home screen ads do not trigger exploration.	76%	Strongly Validated
What are banner frustrations?	Reorders and search listings are not random.	64%	Validated
What behaviors to achieve?	Users actively seek cross-category discovery.	83%	Strongly Validated
What unmet needs emerge?	A post-checkout append window that waives fees.	76%	Strongly Validated

Product Backlog Prioritization: Why Post-Checkout Wins

Evaluating Solutions to the Discovery Gap

To drive category discovery, we evaluated three options using the RICE framework:

- 1. **Smarter Catalog Filters:** High reach, but low impact because users bypass the catalog page entirely to check out.
- 2. **Crate Digger AI Chat Agent:** High impact, but high effort to build and integrate without slowing transaction speed.
- 3. **Post-Checkout Add-on Window:** Wins because it targets the user at the peak intent checkout moment and eliminates the fee barrier. Simple single-rider delivery makes it low effort.

Product Option	Reach	Impact	Conf	Effort	Score
Smarter Filters	80%	1.0	80%	1.5	42.6
Crate Digger Chat	40%	2.0	70%	3.0	18.6
Post-Checkout Window	80%	2.0	80%	1.2	106.6

One Prioritized Solution: The AI-Driven Discovery Engine

THE HERO: Eliminating Surcharge Friction via AI Recommendations

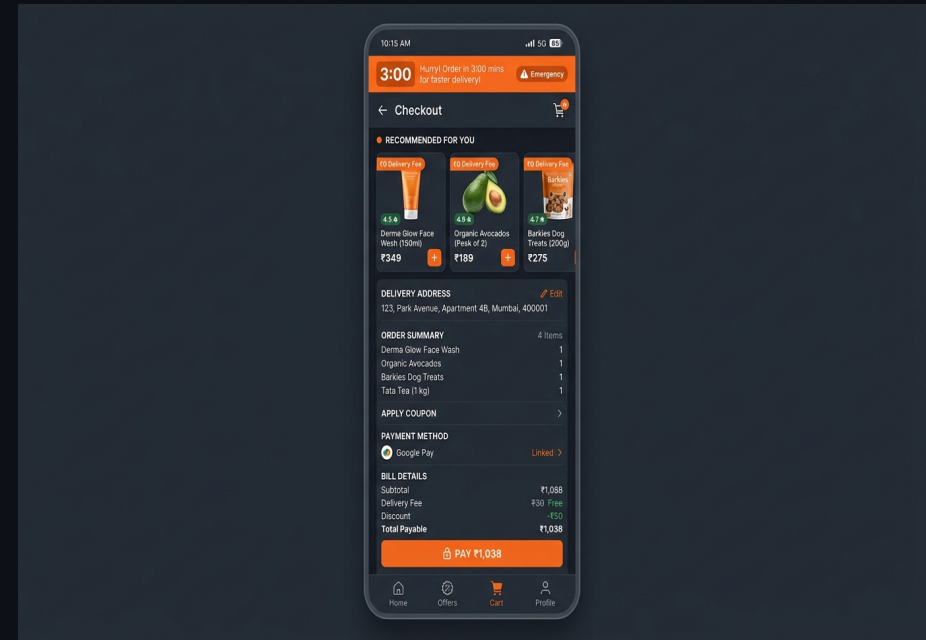
The AI Discovery Engine solves the category exploration gap by introducing a 3-minute post-checkout add-on window. It is built to create a frictionless discovery sandbox:

- **Immediate Intent:** Triggers recommendations right after payment completion, intercepting the customer when they are highly engaged.
- **Zero Surcharge Sandbox:** Waives the ₹35 delivery and packaging fee on recommendations, encouraging low-risk exploratory purchases.
- **ML-Powered Target Catalog:** Shows personalized items from high-margin categories (personal care, pet supplies, organic fruits) based on semantic customer sentiment analyses, instead of repeating past order histories.

The Deployed MVP: 3-Minute Post-Checkout Add-On Window

Interactive Deployed Checkout Sandbox

- **Checkout Simulator:** Clicking check out on everyday dairy essentials simulates payment completion and triggers the active 3-minute window.
- **Dynamic Countdown Banner:** Displays an urgent timer counting down the dark store's packaging cycle.
- **Fee-Free Add-on Carousel:** Displays organic avocados, dog treats, and face washes with ₹0 delivery fee. Selecting an item dynamically updates checkout totals.
- **Try the Deployed MVP:** <https://aashritamalviya1999.github.io/swiggy-instamart-discovery/>



MVP Phased Operational Rollout Strategy

Phased Rollout Stages:

- **Phase 1: Shadow Mode (Weeks 1-2):** Track background behavior. Measure add-on clicks post-checkout without showing UI. Check matching dark store pack cycles.
- **Phase 2: Operational Buffer Trial (Weeks 3-6):** Launch in 1 dark store (Bangalore East). Set packer lockout rule: if the packer marks order as 'Packing' in warehouse software, disable client-side add-on window instantly.
- **Phase 3: Tiered City Expansion (Weeks 7-12):** Expand to 10 dark stores. Integrate with Swiggy One membership.

Operational Lock-out & Rider Safety Gates:

1. **Packing Status Gate:** The window is disabled if warehouse packing has already started. This prevents packers from re-opening sealed packages.
2. **Weight Capacity Gate:** Limit add-on catalog to items <2kg. Heavy items are blocked to keep bikes within safety weight thresholds.
3. **Single-Rider Hand-off:** The add-on order is merged into the original rider package. No secondary rider is assigned.

Phased GTM Pilot & Guardrail Success Metrics

North Star & Leading Indicators:

- **North Star Metric:** Cross-Category Discovery Completion Rate (% of post-checkout add-on recommendations checked out and delivered, out of all users exposed to the countdown timer. Target: 35% completion in 30 days).
- **Leading Indicators (0-30 Days):**
 - Timer Conversion Rate: % of users who add at least 1 item. Target: >25% in week 2.
 - Platform Split Churn Delta: Shift in multi-app usage. Target: -5% churn to Zepto.

SLA Operational Guardrail Metrics:

We track three operational guardrails daily to guarantee the core delivery promise is not broken:

- **Rider Wait Time Delta:** Target: <+30 seconds. Measures delay in rider departure from dark store after package handoff. If delta exceeds 45s, decrease add-on window to 2 minutes.
- **Packer Picking SLA Delta:** Target: <+15 seconds. Measures packer time to add append item to bag.
- **Delivery SLA Breach Rate:** Target: 0%. Percentage of orders delayed beyond Swiggy's delivery promise.

Go-To-Market Operations & Risk Mitigation Matrix

Risk Event	Likelihood	Impact	Packer Mitigation Strategy
Rider delay due to late cart additions	Medium	High	Packer lockout gate: Disable the client add-on option as soon as picker terminal registers order packing commenced.
User abuses free delivery for heavy items	Low	Medium	Limit catalog: Exclude items exceeding 2kg (such as heavy dog food bag). Limit dimensions to fit bike boxes.
Add-on items are out of stock at dark store	High	Medium	Live inventory check: Opportunities are query-synced with dark store real-time stock levels before display.

Target User Cohorts & Persona Deep-Dive

Persona	Behavior Pattern	Core Frustration	Job to be Done	Unmet Need
Rohan, 29 Software Eng Bangalore	Orders milk daily via reorders. Bypasses homepage.	Delivery fee surcharges on small single additions.	Test new categories with zero fee friction.	Post-checkout no-fee append window.
Meera, 42 Homemaker Mumbai	Buys vegetables and grocery items occasionally.	Bad fresh produce batches; wallet refunds.	Buy produce with complete freshness trust.	Visual expiration & freshness metrics.
Priya, 35 Pet/Baby Parent Pune	Buys premium baby wipes and dog food bags.	App lacks diaper size charts and reviews.	Validate specifications before buying.	Detailed specifications charts inside app.

Technical Appendix A: Database & Scraper Ingestion Engine

SQLite Relational Database & Scraper Adapters

- **Schema Structure:** Structured `reviews` database stores raw texts, cleaned values, sentiments, and classifications. The `opportunities` table links directly to dashboard prioritizations.
- **Ingestion Adapters:** Multi-platform Python scraper modules ingest text from Play Store APIs, RSS feeds, and Reddit feeds daily.
- **Cleaning Filters:** Sanitizes emojis, punctuation, and translates mixed Hinglish comments into English database labels for standardization.



Technical Appendix B: 7-Agent Pipeline & Text Vector Clustering

7-Agent Ingestion Orchestration & K-Means Model

- **Sequential Agents:** Pipeline orchestrator coordinates cleaner, translator, clusterer, opportunities scorer, and exporter. This encapsulates all processing logic.
- **TF-IDF Vectorization:** Text fields are converted into a numeric matrix of TF-IDF word importances (max 500 features).
- **Clustering Centroids:** K-Means (K=4) labels reviews based on centroid distances, auto-grouping user friction into Fees, Freshness, and UI noise subthemes.

